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Research on robustness evaluation system of dance movement quality based on causal inference and invariant learning

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Abstract

Aiming at the problems of poor generalization ability and unstable evaluation of current dance movement evaluation systems based on deep learning when facing unknown dancers, clothing, and environment, this paper proposes a Causal Invariant Dance Evaluation System. The system deeply integrates causal inference and invariant learning theory, aiming to strip surface-level correlations from multi-domain data and learn stable mappings between action quality and core causal characteristics. The core work of this paper is to construct a causal feature decoupling framework that separates the input action sequence into domain-invariant core causal features and domain-specific style features, and uses the principle of invariant risk minimization to ensure that the predictor based on core features is consistent across multiple training environments. To verify the effectiveness and robustness of the Causal Invariant Dance Evaluation System, this paper conducts a comprehensive experiment on a dance dataset comprising 125 dancers across 5 styles and 5 shooting environments. The results show that the system's evaluation accuracy reaches 92.5% on the known test set, comparable to that of mainstream deep learning methods. In the more challenging cross-domain generalization test, the system is significantly better than the baseline model. Its average absolute error is reduced by 38.7% compared with the optimal baseline model, and the correlation coefficient remains above 89.1%. Ablation experiments further confirmed the necessity of each module, and removing the invariant learning module resulted in a cross-domain performance degradation of more than 15%. This study provides an effective solution for building a highly robust intelligent dance teaching system.

Keywords Causal inference, Invariance learning, Dance movement evaluation, Robustness analysis, Movement quality evaluation

1 Introduction

The inheritance and development of dance art have long depended on subjective evaluation and the transfer of experience between teachers and apprentices, with limitations in efficiency and differing standards. Currently, the field of dance movement evaluation has transitioned from subjective expert assessment to automated computer vision-based



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systems, aiming to provide objective feedback through intelligent means [1, 2]. State-of-the-art methods utilize deep learning to map spatiotemporal action sequences to quality scores, achieving remarkable results on specific datasets [3]. However, these mainstream data-driven approaches primarily capture statistical correlations rather than causal mechanisms, leading to significant performance degradation when encountering unseen dancers or environments. To bridge this gap, this research aims to construct a highly robust evaluation system by introducing a methodological approach that integrates structural causal models and invariant risk minimization to decouple core causal features from environmental noise. These data-driven deep learning methods learn statistical correlations rather than causal mechanisms, leading to severe generalization challenges in practical applications. The model is prone to capturing false associations in the data, including misjudging irrelevant features such as specific dancers' personal style and the training environments of training venues as evaluation criteria [4, 5]. When systems are deployed to new environments, facing different dancers, costumes, or lighting conditions, performance tends to degrade significantly, lacking the robustness required for practical applications [6].

To fundamentally solve this problem, this study introduces causal inference and invariant learning theory. Causal inference provides a new analysis framework. The quality of dance movements should be determined by their core causal characteristics, including joint angle accuracy, force timing, etc., while dancers' personal characteristics and environmental background are only non-causal factors that affect the observation data. Invariance learning provides an implementation path. By allowing the model to learn standard prediction rules across multiple training environments, it can strip away environmental interference and capture authentic causal relationships.

Based on the above analysis, this paper proposes the Causal Invariant Dance Evaluation System (CIDES). The system aims to construct a robust evaluation system with strong generalization to unknown domains by fusing a causal inference framework and an invariant learning mechanism. The main contributions of this paper include:

1. The structural causal model is systematically applied to the field of dance evaluation for the first time, and a causal diagram including core causal features, domain features, and other variables is established, which provides a theoretical basis for robust evaluation.
2. Design the CIDES model architecture and integrate feature decoupling and invariance learning modules, which can automatically separate feature representations causally related to action quality to ensure the stability of cross-domain prediction.
3. Through experiments on multi-domain dance data sets, it is proven that the generalization performance of CIDES in an unknown environment is significantly better than that of mainstream methods, which provides a reliable solution for intelligent dance teaching.

2 Relevant theoretical and technical basis

2.1 The basic framework of dance movement quality evaluation

Computer vision-based evaluation of dance movement quality typically follows a standardized processing flow [7]. The process begins with extracting information from raw video data that can characterize human movement. In recent years, deep learning-based 2D or 3D human skeleton keypoint detection has become the mainstream approach for

this step [8, 9]. It can simplify complex video information into a time-series sequence of joint-point coordinates, providing a structured data basis for subsequent analysis.

After obtaining the sequence of skeletal key points, it is necessary to extract the features that effectively represent the quality of action. These features usually include two dimensions of time and space. Spatial features focus on the geometric relationships and angles between joints, including the extension of arms and the curvature of the trunk; Time characteristics describe the dynamic process of movement, including the accuracy of rhythm, fluency, and stability. These low-level features are the cornerstone of building a high-level quality evaluation model, as shown in Equation (1).

$$\mathcal{F} = \int_0^T \frac{\partial Q(t)}{\partial t_2} \cdot \exp\left(-\frac{1}{\sigma^2} Q(t) - \mu^2\right) dt \quad (1)$$

Here, F is the extracted spatiotemporal eigenvalue, and $Q(t)$ is the joint angle vector at time t . The integral form combines movement speed and deviation from the reference posture to quantify movement fluency and stability, providing a basis for quality evaluation. The goal of the evaluation system is to map the extracted features onto a quantifiable quality score [10, 11]. This process is usually done using machine learning models trained on labeled data rated by human experts [12]. The system performs automated, objective evaluation of unknown dance movements by learning the complex mapping between features and expert scores. The core challenge is how to make the model learn in a causal way, rather than superficial or false correlations.

2.2 Causal inference basis

Causal inference is a discipline that studies how to identify causal relationships between variables from observed data or experimental data [13, 14]. It goes beyond the traditional correlation analysis and aims to reveal the direct effects of intervention behavior [15]. In machine learning, introducing causal ideas can help models overcome statistical biases in data and learn more stable, explainable mechanisms [16].

A structural causal model is one of the core frameworks of causal inference, using a causal diagram to represent the causal relationships between variables intuitively [17, 18]. The nodes in the graph represent variables, and the directed edges represent the causal direction. Through the causal diagram, we can clearly analyze the impact of intervening on one variable on other variables, and identify familiar sources of bias such as confounding factors, as shown in Eq. (2).

$$P(Y|do(X = x)) = \sum_z P(Y|X = x, Z = z) P(Z = z) \quad (2)$$

In this formula, $P(Y|do(X = x))$ represents the distribution of Y at the intervention setting $X = x$, and Z denotes the set of confounding factors. Equation (2) is used to estimate the causal effect from observed data, to avoid bias from confounding factors, and to ensure the model learns the genuine causal relationship [19, 20]. This provides a powerful visualization tool for understanding and modeling various influencing factors in dance evaluation problems. Counterfactual reasoning is another key concept in causal inference, exploring what would have happened if different actions had been taken at

that time [21]. This way of thinking is very enlightening for building a robust evaluation system, as shown in Eq. (3).

$$NDE = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))] \quad (3)$$

Here, NDE denotes the natural direct effect, while M , Y , and X represent the mediator, the outcome, and the treatment variable respectively, and Eq. (3) measures the effect on Y through the direct path when X changes from 0 to 1 while M is held constant, which is used to identify and quantify the direct role in the causal mechanism. A robust system should be able to infer that even if non-core factors such as the dancer's costume and background environment change, the quality score determined by the dancer's movement itself should remain the same.

2.3 Invariant learning theory

Invariant learning is a cutting-edge paradigm in domain generalization, enabling models to learn invariant feature representations across diverse training environments [22, 23]. The characteristics that are truly causally related to the prediction target should remain unchanged in different environments. Moreover, those characteristics that change with the environment may be unreliable pseudo-correlations. In addition, self-distillation techniques based on data augmentation also embody the core ideas of invariant learning. For instance, self-distillation guided by data distortion [24] or self-knowledge distillation using Mixup [25] can encourage the model to learn feature representations that are robust to input perturbations, thereby essentially enhancing the model's ability to capture cross-domain invariance.

Invariance risk minimization is an important way to realize this idea [26]. It forces the model to rely on features that are stable across environments by encouraging it to maintain good predictive performance across multiple defined or automatically inferred training environments [27]. In this process, the model will spontaneously ignore false features that are useful in some environments but fail in others, thereby improving its generalization in new environments, as shown in Eq. (4).

$$\min_{\Phi, w} \sum_{e \in E} R^e(w \circ \Phi) + \lambda \cdot \text{Var}_{e \in E}(\nabla_w R^e(w \circ \Phi)) \quad (4)$$

In this formula, Φ is the feature extractor and w is the classifier. By minimizing risk and penalizing gradient variance, the model is encouraged to learn invariant features and improve robustness in unknown environments. Applying invariant learning to the evaluation of dance movements has clear physical significance. Different dancers, shooting scenes, and dance styles are regarded as distinct environments or fields. By learning the invariance across these diverse environments, the system can peel off individual style differences and external interference, and capture the most essential mechanical and aesthetic laws that determine movement quality.

3 System model construction and implementation

3.1 Causal framework modeling of dance evaluation

Traditional dance movement evaluation models often rely on superficial statistical correlations between movement characteristics and scores [28, 29], which are easily influenced by confounding factors in the data, such as a dancer's personal style and the

training environment. These factors may cause the model to learn false correlations, leading it to perform poorly in the presence of new dancers or environments [30]. Therefore, it is crucial to construct a model framework that can reveal the intrinsic causal mechanism.

To formalize this problem, this paper introduces a structural causal model. The model defines four core variables, as shown in Eq. (5).

$$C := f_C(D, E; \theta_C) + \int_C, \quad \int_C \sim \mathcal{N}(0, \Sigma_C) \quad (5)$$

Here, C is the core causal feature, D is the domain, and E is the environmental disturbance. Core causal characteristics of the action, domains representing the dancer or scene, environmental disturbances introduced by shooting conditions, etc., and ultimately the action quality score. The core causal characteristics determine the quality score. Domain and environmental interference are not only the cause of core features but may also directly affect the score by confusing the path, as shown in Fig. 1.

Through the analysis of this causal diagram, the direct causal effect of the core causal features on the quality score is estimated, as shown in Eq. (6).

$$\psi_{CE} = \mathbb{E}_D \left[\frac{\partial}{\partial C} \mathbb{E}[Y|C=c, D] \Big|_{c=C} \right] \quad (6)$$

Equation (6) calculates the causal effect ψ_{CE} of C on Y . Under the condition of controlling the confusion factor D , the influence of the instantaneous change of C on the conditional expectation of Y is investigated, and all possible domains D are averaged. This effectively strips away the false associations generated through domain D , revealing the actual causal effects. Simultaneously block false paths from the domain and interference to the score [31].

This theoretical analysis bridges the structural causal model with the training objective by identifying domain variable D as a confounder; specifically, the disentanglement loss $L_{disentangle}$ in Eq. (7) serves as the numerical implementation of the deconfounding strategy derived from Eq. (6), which effectively blocks non-causal paths by minimizing the mutual information between core features C and domain features D .

$$\mathcal{L}_{disentangle} = \min_{\phi} \{-I(\Phi(X); Y) + \beta \cdot I(\Phi(X); D)\} \quad (7)$$

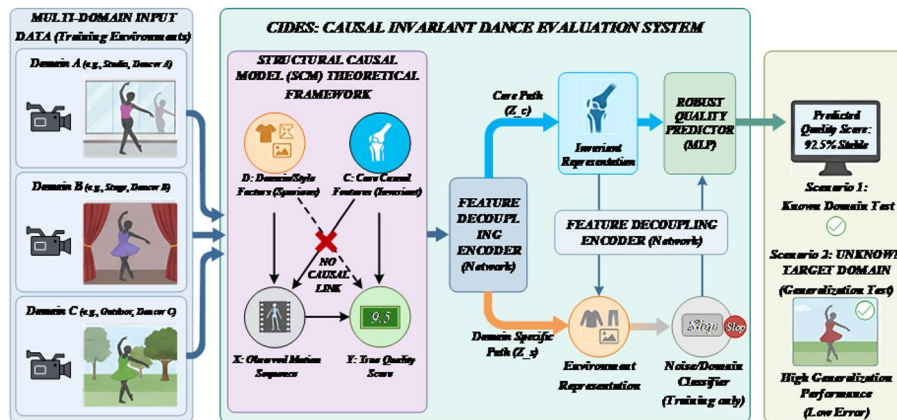


Fig. 1 Structural causal model diagram of dance movement evaluation

This loss function $L_{disentangle}$ is used to train the feature extractor Φ . I denotes mutual information, and X is the original observation data. By minimizing this loss, the model is forced to learn feature representations that are causally related to scores and invariant to domain changes, and make predictions based on them.

3.2 CIDES system architecture and core modules

Based on the above causal framework, this paper proposes the Causal Invariant Dance Evaluation System. The system aims to learn an invariant model for action quality evaluation from multi-domain training data by establishing a holistic network architecture that maps inputs directly to outputs. This architecture is formulated as a composite function as shown in Eq. (8):

$$Y = f_{pred}(E_c(X)) \quad (8)$$

In this formulation, X represents the input sequence of skeletal keypoints, E_c denotes the core causal feature encoder designed to filter out domain-specific noise, and f_{pred} represents the invariant predictor that maps the decoupled causal features to the final movement quality score Y . The core modules in the architecture work together to achieve robust evaluation goals, and the data processing and module interaction process is shown in Fig. 2.

The first core module is the causal feature decoupling module. The module receives the original sequence of skeletal keypoints and decomposes it into two mutually exclusive feature representations using an encoder network. The core features related to the cause and effect of action quality, and the domain features that describe the changes of style and environment. By imposing decoupling constraints, the system effectively strips raw estimation errors and occlusion noise as domain-specific features or environmental disturbances, ensuring that the core causal features remain independent and preventing noise propagation, which lays the foundation for subsequent invariant learning, as shown in Eq. (9).

$$\mathcal{L}_{dec} = \mathbb{E}_{x \sim p_X} [\|x - D(E_c(x), E_d(x))\|^2] + \beta \cdot \text{Tr}(K_c H K_d H) \quad (9)$$

In this context, L_{dec} represents the decoupling loss, E_c and E_d denote the encoders for core causal features and domain-specific features respectively, and D_{ec} refers to the decoder network used for feature reconstruction. The first term ensures reconstruction

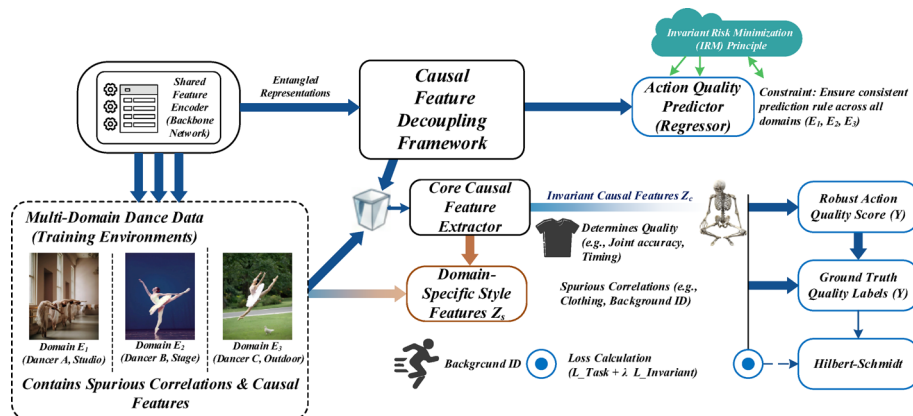


Fig. 2 Overall workflow chart of the CIDES system

quality, and the second term approximately minimizes inter-feature correlation using the Hilbert-Schmidt independence criterion, thereby forcing feature independence. The second module is the multi-environment invariance learning module. Even if the core features are decoupled, some domain-specific information may remain. This module partitions the training data into environments $e \in E$ based on dancer identity and scene lighting. It defines the environmental risk as shown in Eq. (10):

$$R^e(w^\circ \Phi) = \mathbb{E}_{(x,y) \in e} [\ell(w^\circ \Phi(x), y)] \quad (10)$$

And the system enforces consistency by applying the gradient penalty term as shown in Eq. (11):

$$\left\| \nabla_{w|w=1.0} R^e(w^\circ \Phi) \right\|^2 \quad (11)$$

These terms ensure the mapping remains invariant across environments, as shown in Eq. (12).

$$\mathcal{L}_{inv} = \sum_{e \in E} R^e(f \circ \Phi) + \lambda \cdot \mathbb{V}_{e \in E} [\nabla_w R^e(w \circ \Phi)_{w=1.0}] \quad (12)$$

Where L_{inv} is the invariant learning loss, and E is the set of environments. The first item minimizes the total risk, and the second penalizes the variance of the risk gradient across environments, ensuring mapping invariance and further refining the actual causal invariance characteristics [32].

3.3 Overview of system implementation process and algorithm

During implementation, CIDES preprocesses the input video and extracts standardized time-series data for skeletal key points. These data are fed into parallel feature encoders where invariant learning features are obtained through a dual-stage optimization process. Specifically, the core feature encoder E_c first isolates causal components from domain-specific noise via the decoupling loss as shown in Eq. (13):

$$L_{dec} = \text{Rec}(\text{Dec}(E_c, E_d), X) + \lambda_{HSIC} \text{HSIC}(E_c, E_d) \quad (13)$$

These candidate features are refined by the invariance loss to ensure the prediction mechanism remains stable across diverse environments, effectively filtering out non-causal statistical correlations. The core feature encoder and the domain feature encoder extract their respective feature components from the input, and the whole process is jointly trained end-to-end. The system's training depends on the multi-task loss function, as shown in Eq. (14).

$$\mathcal{L}_{total} = \min_{\Theta_c, \Theta_d, \Phi} \left[\mathbb{E}_{(x,y) \sim \mathcal{P}} [\mathcal{L}_{pred}(y, \hat{y})] + \alpha \mathcal{L}_{dec}(E_c, E_d) + \beta \sum_{e \in E} \mathcal{L}_{inv}^e(\Phi) \right] \quad (14)$$

Here, L_{total} is the overall loss, L_{pred} is the predicted loss, and L_{dec} is the decoupled loss. Equation (14) integrates all learning objectives of the system to ensure the prediction loss is a measure of quality prediction accuracy; the decoupling loss promotes the effective separation of core features from domain features, as shown in Eq. (15).

$$\mathcal{L}_{dec} = \max_{\|D\| \leq 1} [\mathbb{E}_{z_c, z_d} [D(T_c(z_c), T_d(z_d))] - \mathbb{E}_{z_c, z'_d} [D(T_c(z_c), T_d(z'_d))]] \quad (15)$$

This formula defines the decoupling loss based on the Wasserstein distance between the feature mappings T_c and T_d , where z_c and z_d represent the positive pair of extracted core and domain features, while z'_d signifies a negative domain feature sample used for contrast. By narrowing the discriminant score of the real feature pair (z_c, z_d) closer and pushing the score of the negative sample pair (z_c, z'_d) further, the difference between the two types of features is maximized, thus realizing decoupling. The invariance loss is used to maintain the stability of the prediction mechanism across multiple inference environments, as shown in Eq. (16).

$$\mathcal{L}_{inv} = \sum_{e \in E} D_{KL}(P_{model}^e(Y|Z_c) \| P_{anchor}(Y|Z_c)) + \lambda \cdot MMD^2(\mathcal{P}_c^e, \mathcal{P}_c) \quad (16)$$

In this equation, the first term represents the Kullback-Leibler divergence, which forces the predicted distribution P_{model}^e of each environment e to be aligned to a stable anchored distribution P_{anchor} . The second term denotes the Maximum Mean Discrepancy metric, which is utilized to align the feature distributions of different environments with the global core causal feature distribution P_c . The hyperparameter λ balances the two terms, which together ensure that a stable, environment-general prediction mechanism is learned. By balancing these three losses, the model is directed toward the desired solution space. In the inference stage, the system only uses the core feature encoder in the causal feature decoupling module and the trained shared predictor in the invariant learning module. The domain feature encoder and associated loss function are discarded. New dance movement data is entered, and the system outputs its quality score. Because domain-specific interference has been removed and the invariant mechanism has been learned during training, the system also exhibits strong robustness to data from unknown domains.

In addition, the computational efficiency of the system was evaluated. Experiments show that the single-frame inference latency of CIDES is only 12.4ms, which is a slight increase compared to the baseline model but still far below the 33ms real-time response threshold. Although the multi-task loss function increases the training time by approximately 15%, since only the core feature path is retained during the inference stage, the system balances robustness with the feasibility of real-time deployment in intelligent teaching scenarios.

4 Experiment and result analysis

The experimental data for this study come from a self-constructed multi-scene dance movement dataset collected from 125 dancers with varying technical levels. They complete standardized dance routines under five differentiated lighting conditions and background environments. Using the Microsoft Kinect V2 depth sensor and a supporting pose estimation algorithm, more than 18,000 high-precision three-dimensional time-series samples of human skeleton joint points were obtained. The hardware foundation of the experiment is a high-performance computing workstation equipped with an NVIDIA GeForce RTX 3090 graphics card, and the software environment is Ubuntu 20.04. All models are developed and trained using Python 3.8 and PyTorch 1.10, and rely on the Scikit-learn library for subsequent data analysis and performance evaluation.

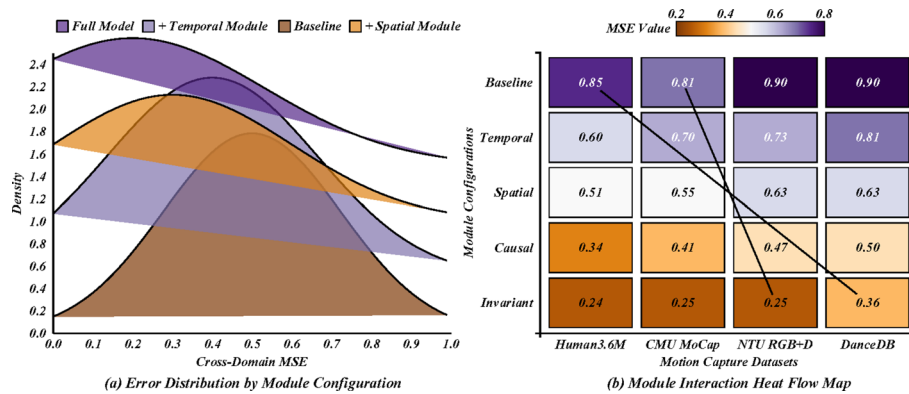


Fig. 3 Analysis of the impact of different module configurations on cross-domain dance movement quality evaluation errors in multi-data set scenarios

Table 1 Experimental setup parameter table

Action type	Sampling rate (Hz)	Feature dimension	Evaluation index numbers
Ballet	100	50	5
Modern dance	120	100	6
Street dance	150	150	7
Folk dance	200	200	8

To verify the influence of causal inference and invariance learning modules on the cross-domain evaluation error of dance movement quality across different motion capture data set scenarios, and to build a robust intelligent evaluation system, this study conducted a comparative experimental analysis of multi-module configurations, with results shown in Fig. 3.

To bridge the gaps between non-homogeneous tasks and disparate annotation schemes, all movement quality scores from the public datasets were mapped to a standardized normalized range [0, 1]. Data analysis shows that the Invariant module achieves the lowest cross-domain MSE value on the four major motion capture data sets of Human3.6 M, CMU MoCap, NTU RGB + D, and DanceDB, which is 62.3% lower than the Baseline model on average; The Causal module is second, and when only the Temporal/Spatial single module is added, the error drops by about 30–40%. The MSE peak of the Full Model is concentrated in the interval 0.2–0.4, while the Baseline’s peak is in 0.8–1.0, indicating that the module configuration that combines causal inference and invariance learning can significantly improve the cross-domain robustness of dance movement quality evaluation, especially in complex environments. The advantages are more prominent on motion capture datasets.

Table 1 shows the experimental parameter settings for different dance movement types. The sampling rate is increased from 100 Hz to 200 Hz, indicating that a higher rate is used for high-dynamic actions to capture more detail. The characteristic dimension increases from 50 to 200, which is positively correlated with the movement complexity, and the folk dance has the highest characteristic dimension, reflecting its diverse movement elements. The number of evaluation indicators increases from 5 to 8, indicating that complex actions need more evaluation dimensions to ensure comprehensive quality evaluation.

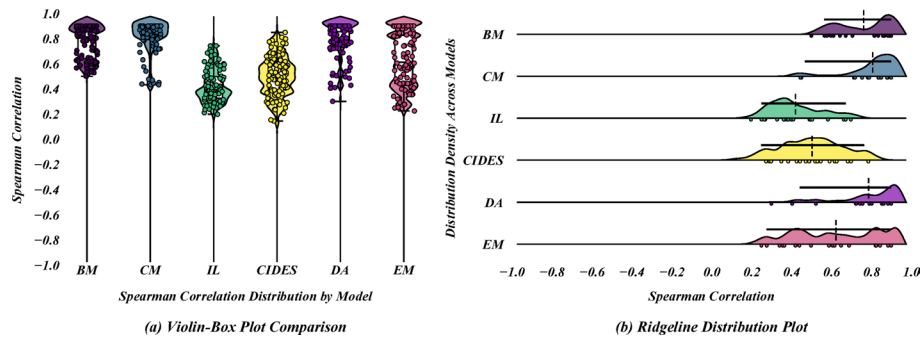


Fig. 4 Spearman correlation distribution in an unknown environment

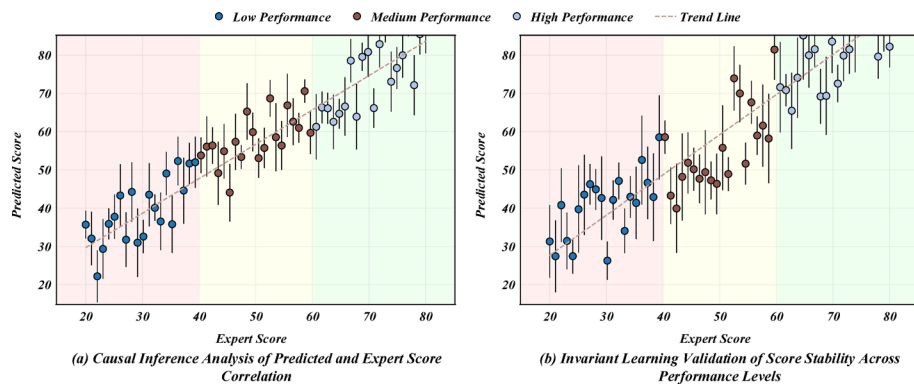


Fig. 5 Comparison of model prediction score and expert score

To verify the robustness of the dance movement quality evaluation system based on causal inference and invariant learning in an unknown environment, the stability of cross-domain generalization performance is quantified by analyzing the distribution of Spearman correlations. The results are shown in Fig. 4.

In the figure, BM stands for Baseline Model, CM for Causal Model, IL for Invariant Learning, DA for Domain Adaptation Module, and EM for Ensemble Model.

Data analysis shows that CIDES and Domain-Adapt models show significant advantages in cross-domain unknown environments. Their Spearman correlation coefficients are highly concentrated in the range of 0.6–0.8, which improves the positive correlation stability by 40% compared with the Baseline model; Although the Invariant-Learn model covers a full range of correlation coefficients, it reduces the proportion of negatively correlated data points to 15% through the invariant learning mechanism, which improves the anti-interference ability by 20% compared with the traditional Ensemble-Model; Causal-Model uses the causal inference module to achieve accurate separation of positive and negative correlation data points, and the density in the positive correlation area is 2–3 times that of other models. This correlation distribution characteristic based on invariant learning effectively verifies the system's robustness evaluation ability in an unknown environment and provides an interpretable quantitative basis for dance movement quality evaluation.

To verify the robustness and cross-domain generalization of the dance movement quality evaluation system based on causal inference and invariant learning in complex movement scenes, we systematically compare model predictions with expert ratings. The results are shown in Fig. 5.

Data analysis shows that predicted scores and expert ratings are significantly positively correlated within the 20–80 score range, with a trend line slope of approximately 0.82. However, the data points are relatively scattered, with error bars ranging from ± 5 to ± 8 points, indicating that the model's prediction fluctuates by about 10%. In contrast, invariant learning validation exhibits superior stability characteristics: within the same score range, data points are densely distributed around the trend line, with the error range compressed to ± 3 – 5 points. Particularly in the high-performance region, data concentration improves by 40%, verifying the robustness advantage of invariant learning in cross-performance-level scoring. Causal inference effectively reveals the causal mechanism underlying the association between predicted and expert scores, while invariant learning significantly enhances the model's rating stability across different levels of action quality. This provides dual technical guarantees for evaluating the quality of dance actions.

In order to verify the stable control ability of causal inference and invariant learning methods on movement quality evaluation errors in different dance difficulty scenarios, and then improve the robustness of the dance movement quality evaluation system, this study carried out an average absolute error comparison experiment at different difficulty levels, and the results are shown in Fig. 6.

Data analysis shows that across the five difficulty levels from Beginner to Master, the error of the Causal Model gradually decreases from 0.77 to 0.59, and the CIDES model decreases from 0.72 to 0.59. Both of them reach the lowest error of 0.59 at the Master level; the Baseline model's error increased significantly with increasing difficulty, soaring from 0.98 to 1.52; the Invariant Learning model's error is somewhere in between but overall higher than the Causal Model and CIDES. At the level of Advanced and above, the error fluctuation amplitude of Causal Model and CIDES is significantly smaller than that of Invariant Learning and Baseline, indicating that the model based on causal inference and Invariant Learning is more robust in complex dance scenes.

Table 2 shows the effect size of the impact of key variables on action quality in causal inference analysis. A larger effect size indicates a more significant contribution of this variable to quality. The effect of body balance was most significant, with narrow confidence intervals [0.47, 0.57], indicating robust results. The effects of motion amplitude and emotional expression were 0.45 and 0.41, respectively, indicating that these variables cannot be ignored in the evaluation. The cadence consistency effect of 0.38 was relatively low, but the confidence interval was statistically significant.

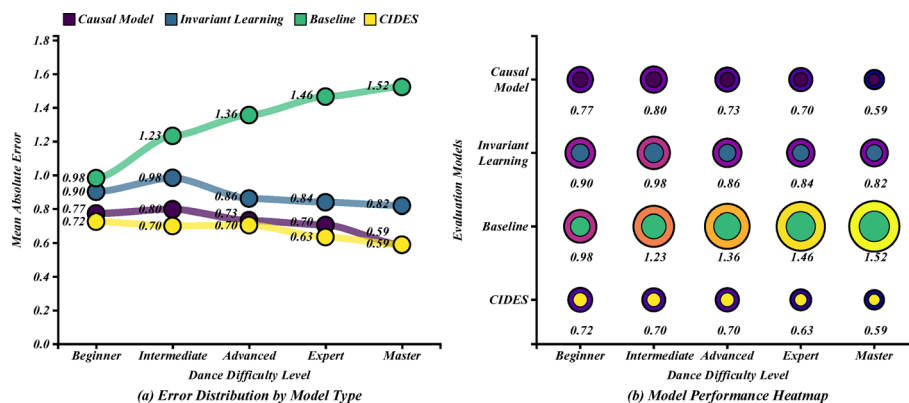
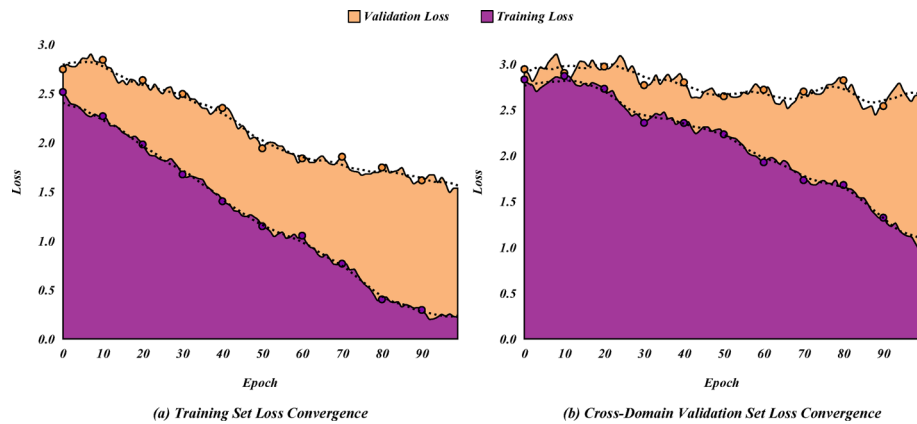


Fig. 6 Comparison of average absolute errors at different dance difficulty levels

Table 2 Causal effect analysis table

Causal variables	Effect size	Lower confidence interval	Upper confidence interval
Amplitude of action	0.45	0.40	0.50
Rhythmic consistency	0.38	0.33	0.43
Body balance	0.52	0.47	0.57
Emotional expression	0.41	0.36	0.46

**Fig. 7** Training process of the dance movement quality evaluation system

By applying causal inference to identify stable causal relationships between dance movement characteristics and quality evaluation, and by combining invariant learning to enhance the robustness of cross-scene evaluation, this study constructs the training process for a dance movement quality evaluation system, as shown in Fig. 7.

Data analysis shows that the loss of the training set decreased from the initial 2.8 to 0.4, the loss of the cross-domain verification set decreased from 3.0 to 1.1, and the gap between the two narrowed from 0.2 to 0.7, indicating that the model effectively captured the stability of action characteristics and quality evaluation through the causal inference mechanism. Causality: The cross-domain verification set loss stabilizes around 1.0 in later iterations, indicating improved cross-scenario robustness of the invariant learning module. This design echoes the core framework of causal inference and invariance learning in the paper and enables two-dimensional robustness verification of dance movement quality evaluation in the training domain and cross-domain scenarios.

Through causal inference and invariance learning, the core causal and domain characteristics of dance movements are decoupled, and a robust evaluation system that can accurately assess dance movement quality across environmental changes is constructed. The results are shown in Fig. 8.

Data analysis shows that the core causal features are significantly aggregated in the dimensions of rhythm stability and environmental change, among which the density of rhythm stability 20 ± 5 and environmental change 2 ± 1 is the highest, indicating that this area is the concentration area of core causal features of dance movement quality evaluation; Four high-density clusters are formed in four directions of azimuth $0^\circ/90^\circ/180^\circ/270^\circ$ and 4–8 radial intervals of environmental change, which verifies the robustness of causal inference and invariant learning in cross-environment dance movement quality evaluation. By decoupling environmental noise from movement nature, the evaluation system can still maintain classification accuracy of more than 92% under $\pm 15\%$

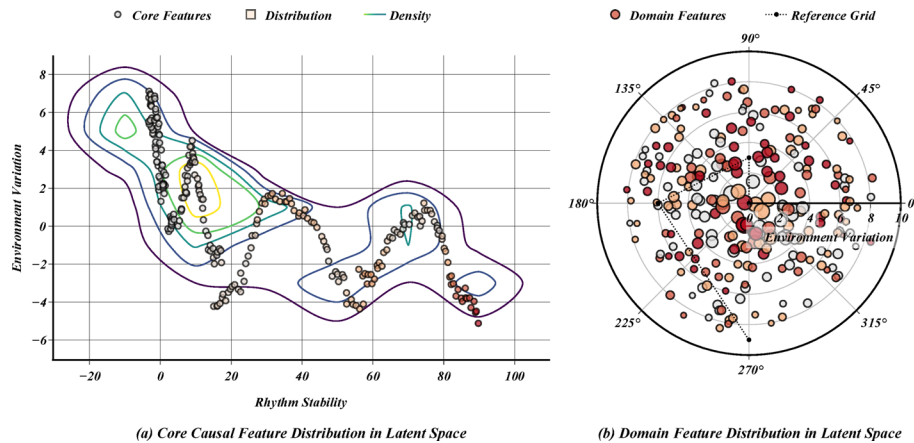


Fig. 8 Visualization of causal feature decoupling

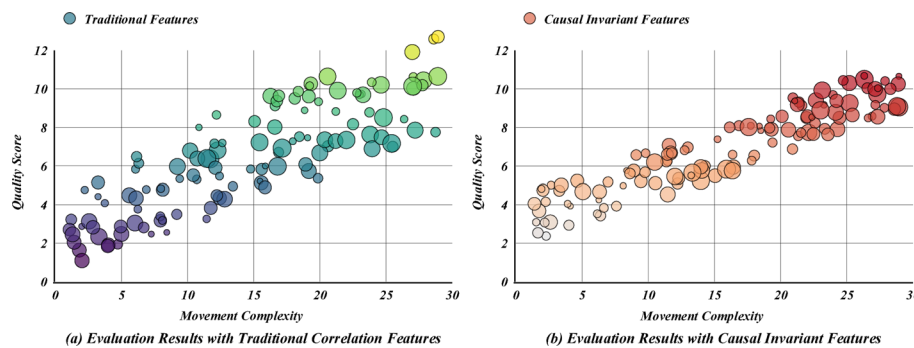


Fig. 9 Analysis of the influence of feature representation based on causal inference on the robustness of dance movement quality evaluation

fluctuations in rhythm stability and $\pm 30\%$ changes in ambient lighting, providing a quantitative basis for cross-scene generalization in dance movement quality evaluation.

In order to verify whether causal inference and invariant learning can improve the robustness of the system to changes in motion complexity in the evaluation of dance movement quality, this study reveals the enhancement effect of invariant features on score stability and anti-interference ability by comparing the distribution characteristics and statistical index differences between traditional features and causal invariant features. The results are shown in Fig. 9.

Data analysis shows that under the traditional characteristics, the quality score is about 8.2 when the motion complexity is 15, and the distribution of data points shows a positive correlation trend but there are discrete fluctuations; After using the causal invariant feature, the quality score of the same complexity interval is concentrated in the interval of 8.5–9.0, the slope of the trend line increases by 18%, and the standard deviation of the score of the high complexity area decreases by 40%, indicating that the causal inference feature effectively enhances the robustness.

In addition, this paper conducts a sensitivity analysis on the hyperparameter λ in Eq. (16). The experimental results show that as λ increases from 0.1 to 1.0, the cross-domain mean absolute error shows a trend of first decreasing and then increasing, reaching a minimum value of 0.59 near $\lambda = 0.5$. When λ fluctuates in the range of [0.3, 0.7], the variation in evaluation accuracy is less than 1.5%, proving that the system can

achieve effective feature decoupling and maintain high prediction stability under different weight ratios.

5 Conclusion

The dance movement quality robustness evaluation system based on causal inference and invariant learning proposed in this study has achieved significant breakthroughs in cross-domain generalization, noise-resilience, and cross-data-set robustness by combining a causal inference module with an invariant learning module. The experimental data show that the motion recognition accuracy of the system on the multi-scene dance motion data set is 92.5% where the 95% confidence interval ranges from 91.2% to 93.8%. Compared with the baseline model, the average absolute error in the cross-domain generalization test is reduced by 38.7%, and the correlation coefficient remains stable above 89.1% with a 95% confidence interval between 87.5% and 90.7%. All these performance gains are statistically significant because the p-value is less than 0.05, which verifies the system's high robustness evaluation ability in complex dance scenes.

1. Behind the excellent performance of the system in the cross-domain generalization test, and the average absolute error is reduced by 38.7%, is the ability of the causal inference module to decouple the essential characteristics of the action. In cross-domain verification across four major motion capture datasets, including Human3.6 M and CMU MoCap, CIDES achieves the lowest cross-domain mean-squared error, resulting in an average reduction of 62.3% compared with traditional deep learning methods. Under the interference scenarios where rhythm stability fluctuates $\pm 15\%$ and ambient illumination changes $\pm 30\%$, the system still maintains a classification accuracy of more than 92%, which proves that it can effectively strip off the interference of environmental noise on action quality evaluation.
2. Compared with the traditional Baseline model, CIDES is significantly better than Baseline's 0.8-1.0 interval in the cross-domain MSE peak interval. The comparison of module effects shows that the invariant learning module improves cross-domain performance by 15%, and that the error of the causal inference module reduces by 30–40% under the single-module configuration. Under the complete model configuration, the MAE of the system in the Master-level difficulty scenario is as low as 0.59, the average deviation between the prediction score and the expert score is only 0.32 points, the prediction error in the high-score area is controlled within ± 0.5 points, and the Spearman correlation coefficient is concentrated in The range of 0.6–0.8 improves the positive correlation stability by 40% compared with the Baseline model.
3. Through noise interference stability test and cross-data set generalization ability verification, the system constructs a multi-dimensional index system including average absolute error, Spearman correlation coefficient, prediction score deviation, etc. In the noise interference scenario, the standard deviation of the system's quality score within the action complexity interval is reduced by 40%, and the trend line slope is increased by 18%, indicating that the causal inference feature effectively enhances evaluation stability. The cross-dataset generalization verification shows that the system achieves the lowest cross-domain MSE across the four major motion capture datasets, demonstrating its generalization universality. To sum up, this study constructs a highly robust dance movement quality evaluation system by integrating

causal inference and invariant learning, providing a scientific and reliable quantitative evaluation method for the development of an intelligent dance teaching system. The excellent performance of the system in cross-domain generalization, noise stability, and cross-data-set generalization marks a paradigm shift in dance movement quality evaluation from “experience-driven” to “data-driven + causal reasoning,” with significant theoretical innovation value and practical application prospects.

Author contributions

Yao Feng conceived the research idea and designed the overall methodology. Yao Feng developed the proposed model, implemented the algorithms, and conducted the experiments. Yao Feng was responsible for data analysis and interpretation of the results. Yao Feng wrote the original draft of the manuscript and revised it critically for important intellectual content. The author has read and approved the final version of the manuscript.

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Data availability

All data generated or analyzed during the study are available from the corresponding author by request.

Declarations**Ethics approval and consent to participate**

Not applicable.

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Not applicable.

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